Baselines and fair comparisons

Learn to understand, design, build, test and explain AI systems responsibly.

Project: Reproducible AI Research Study

THE CONNECTED WEEK

Use a credible baseline. Match the evidence. Respect the task budget.

Explain source confounds, unseen-evaluation leakage and complete-procedure trade-offs with bounded comparisons.

Keep one habit: Baseline contract → source-only counterexample → eight-case picture → leakage/resource probes → condition audit → independent time budget.

MAIN LESSON · SECTION 1

Give the baseline a credible method and comparable information

> Define what is held fixed and what difference the study intends to assess.

Learning goals

I can explain the **mechanism** and terminology of **baseline**s and fair comparisons.

I can apply the stated procedure to a new case with a traceable result.

I can identify an assumption or limitation and design a revealing follow-up.

Retrieve primary outcomes and comparison conditions

Close your notes first. A rough first answer helps us choose the right starting point; it is not a score for your ability.

Without opening the old lesson, explain one key idea from Hypotheses and measurable outcomes. Give an example and a mistake that the idea helps you avoid.

Without opening the old lesson, explain one key idea from Replication and reproduction. Give an example and a mistake that the idea helps you avoid.

Without opening the old lesson, explain one key idea from Reading a research claim. Give an example and a mistake that the idea helps you avoid.

After trying, check the relevant lesson or ask your teacher. Change the part of your explanation that the **evidence** shows was wrong.

Prerequisite and starting response

- Recall Hypotheses and measurable outcomes. Explain why a recorded result needs its **input** conditions and **evidence** source before it can support a claim.

My starting explanation

End goal

Explain **baseline**s and fair comparisons using the records and boundaries of this fictional investigation.

▼ **Earlier wording, alternatives and picture transcripts**

Your learning target

Remember before reading

This week you will practise:

Explain the **mechanism** and terminology of **baseline**s and fair comparisons

Apply the stated procedure to a new case with a traceable result

Identify an assumption or limitation and design a revealing follow-up

Retrieve what you already know

This week's end goal

Define a baseline and the intended comparison before looking at scores

> A weak reference or an information mismatch can exaggerate improvement.

Begin with yesterday's versus today's timetable

Imagine comparing two room helpers while giving one yesterday's timetable and the other today's. A higher score could come from fresher information rather than a better lookup method. Use a credible documented baseline, the same eligible current sources, matched cases and scoring, and a declared resource policy.

If you intentionally compare unequal complete systems, report the unequal conditions rather than claiming the method alone caused the difference.

Define credible fair baselines

Fair **baseline**s use credible procedures on matched **evidence** and resource conditions. A weak arbitrary reference can exaggerate improvement.

Complete source, workload and resource contract

Fair-comparison item	Complete fictional classroom rule
Credible baseline	Manual authoritative lookup or deterministic lookup matching the requested room; not arbitrary random rooms
Matched evidence	Same eligible current sources for a method-focused comparison
Matched workload/scoring	Same fixed case IDs, outcome rubric and evaluation labels kept out of tuning inputs
Resource policy	Declare timing boundary, budgets and what each procedure actually consumes; unknown is not automatically matched
Added main fixture	Same lookup function on clubs V/W/X; only source freshness differs
Current register/truth	V→E, W→B, X→D
Old register	V→C, W→B, X→A
Separate picture	Same eight fixed questions and allowed sources; A 6/8 correct, B 7/8; no test-case tuning
Limits	Constructed counterexample/aggregates, not a model benchmark, measured runtime, stable advantage or significance result

Name supplied and missing evidence

List the given facts and one fact that is not supplied.

▼ **Earlier wording, alternatives and picture transcripts**

Begin by writing the question the procedure is intended to answer. Identify the information already supplied, the information it will produce and the information still missing.

These are constructed classroom examples; they do not require live accounts, private records or an external service. Do not treat a plausible explanation as a measured outcome.

MAIN LESSON · SECTION 3

Trace a score gap produced by source freshness alone

> Use the counterexample to locate a comparison confound rather than diagnose every system.

Original same-source baseline procedure

For a room-reporting study use manual authoritative lookup or a deterministic lookup matching the requested room, not random room names. Match cases, available sources and scoring.

Complete constructed old/current register ledger

Added same-procedure fixture	Current correct room	Old-source lookup	Current-source lookup
Club V	E	C: incorrect	E: correct
Club W	B	B: correct	B: correct
Club X	D	A: incorrect	D: correct
Score on three constructed cases	Same judge and function	1/3	3/3

Added same-procedure fixture	Current correct room	Old-source lookup	Current-source lookup
Align both to current register	Same cases/function/current evidence	3/3	3/3
Meaning	Only information changed in this demonstration	Score gap can occur without a method difference	Not a prediction that real A/B models must tie

State the intended variable and remaining conditions

Before applying the procedure, write its units, denominator or decision boundary where relevant. A partner should be able to repeat each step using the same **input**s.

If a required **input** is absent or violates the stated contract, record that condition explicitly instead of quietly replacing it with a convenient value.

Explain the visual

Point to the **input**, the deciding step and the result in this week's visual. Explain the same step in ordinary words. A picture helps when you can say what each part represents.

▼ Earlier wording, alternatives and picture transcripts

Baselines and fair comparisons — evidence trace

INPUT AND SCOPE

Assistant gets current register; baseline gets an obsolete one. Assistant scores higher.

CHECK OR CALCULATION

For a room-reporting study use manual authoritative lookup or a deterministic lookup matching the requested room, not random room names.

CONCLUSION AND LIMIT

Source freshness is confounded with procedure. Provide both the same eligible current sources or report the unequal comparison without isolating method.

The figure connects this chapter's worked input, declared procedure and result. Read the complete worked reasoning for qualifiers and intermediate steps; the abbreviated labels do not replace that reasoning.

Connect the picture and the explanation

MAIN LESSON · SECTION 4

Predict hidden answer-key exposure before testing

> A revealed test answer is different from permitted task evidence.

Current task and response

- Assistant gets current register; **baseline** gets an obsolete one. Assistant scores higher. Show the intermediate steps and give the conclusion with its scope.

> **Keep the procedure fixed. Explain what changed in the inputs and result.**

Predict giving B the evaluation key

- Change: Give B the answer key before testing.

▼ **Check after predicting**

Prediction: The resulting comparison is invalid as unseen evaluation.

Reason: B would receive privileged information that A did not receive.

▼ **Earlier wording, alternatives and picture transcripts**

Compare a familiar case with a changed case

Change one thing in the pictured case

MAIN LESSON · SECTION 6

Read the observed gain beside the comparison conditions

> Keep source access, unseen labels and resource evidence distinct.

Main information-freshness confound and checked repair

Assistant gets current register; **baseline** gets an obsolete one. Assistant scores higher.

Source freshness is confounded with procedure. Provide both the same eligible current sources or report the unequal comparison without isolating method.

Compare the intermediate **evidence** with the final statement. Explain why each number or decision follows from the stated **input**s.

The conclusion belongs to this scenario and procedure; changing the **input**s, evaluation population or authority can change what it establishes. Retain the raw record so a later revision can be compared honestly.

Separate eight-case result and complete condition ledger

A **baseline** deserves a fair chance. Two classifiers answer the same eight fixed questions with the same allowed sources.

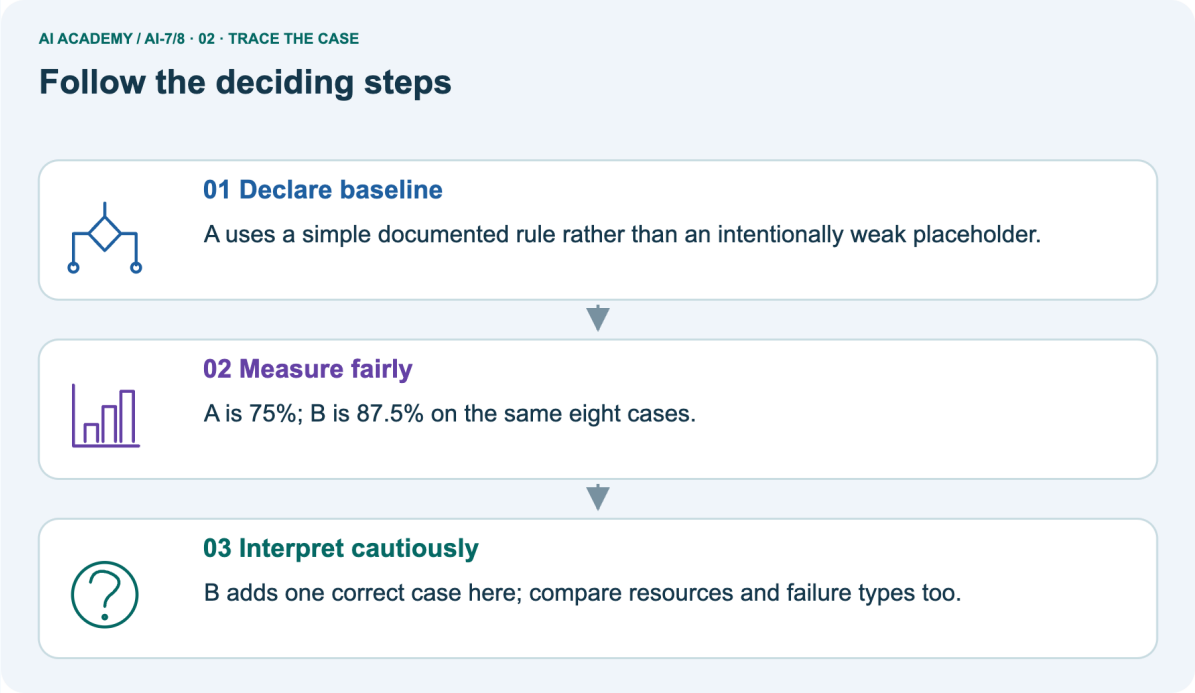
Simple rule A: 6/8 correct. New model B: 7/8 correct. No test case is used to tune either version.

The picture is a separate eight-case baseline/model comparison. A 6/8 =75%; B 7/8 =87.5%; difference 12.5 percentage points and one net additional correct answer. Aggregates do not reveal every paired gain/loss. Cases, allowed sources and scoring are matched in the scenario, with no test-case tuning; resources remain unreported. Giving B the hidden evaluation answer key would break the unseen comparison.

An approved source register that the task permits reading is not the same as exposing held-out judge labels. The supplied aggregate comparison has no statistical analysis or broad-advantage evidence, and should not be reported as one.

Separate eight-case picture	A baseline	B new model	Interpretation
Correct/total	6/8	7/8	Same fixed case denominator
Observed rate	75%	87.5%	B higher on supplied cases
Difference	Reference	+12.5 percentage points	One net additional correct count; paired flips not supplied
Permitted sources/cases	Same declared sources/cases	Same declared sources/cases	Relevant matching conditions
Tuning on test answers	None in original scenario	None in original scenario	Changed answer-key exposure breaks unseen evaluation
Resource/timing evidence	Not supplied	Not supplied	Cannot infer budget eligibility or efficiency

Trace credible baseline, matched scoring and limited gain



Follow the labelled picture with the matching explanation. Open the vector diagram

Read the picture one step at a time

STEP	WHAT HAPPENS AND WHY
1. Declare baseline	A uses a simple documented rule rather than an intentionally weak placeholder.
2. Measure fairly	A is 75%; B is 87.5% on the same eight cases.
3. Interpret cautiously	B adds one correct case here; compare resources and failure types too.

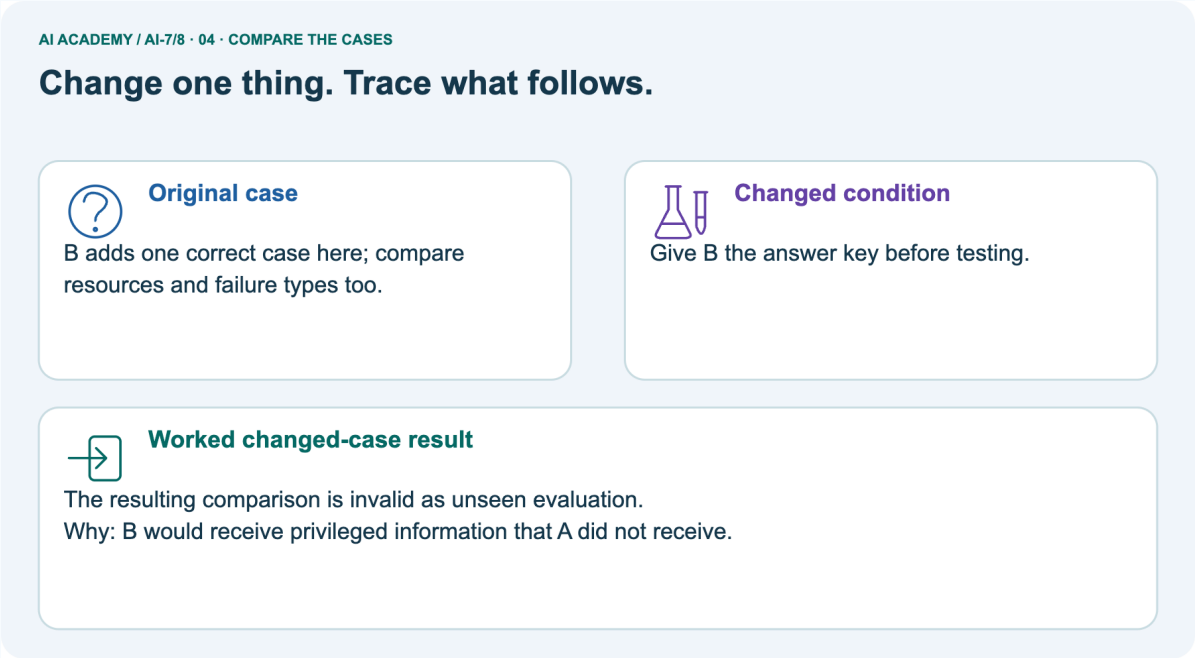
Match each pictured step to the supplied facts. This is a worked teaching case, not an independent assessment.

A **baseline** anchors the claim of improvement. Fair comparison keeps information access, cases and scoring aligned while allowing the declared method difference.

Predict answer-key exposure

- Change one thing: Give B the answer key before testing. Predict the result before checking the explanation. What remains the same?

▼ Check after predicting



Follow the labelled picture with the matching explanation. Open the vector diagram

Explain absent statistical/resource evidence

- These eight-case aggregate results alone establish neither a stable advantage nor a reported statistical-significance result.

- Read the labelled picture: Calculate 6/8 and 7/8 correctness and name the one-case gain. Why is giving B the answer key incompatible with an unseen comparison to the simple **baseline**?

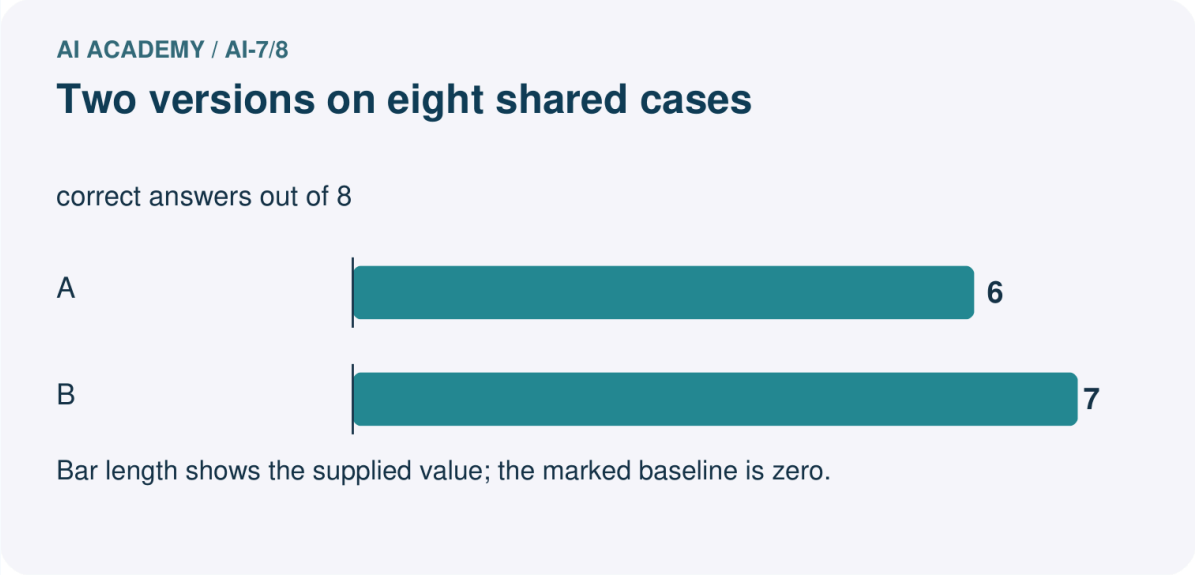
▼ Earlier wording, alternatives and picture transcripts

Learn from the worked case

Cover the result before checking it. Say what is given, choose the procedure, and follow one step at a time. Then uncover the result and explain your first mismatch. Ask why a step is allowed, not only what number it produces.

A pictorial worked case: Baselines and fair comparisons

The facts we will use



Follow the labelled picture with the matching explanation. Open the vector diagram



Follow the labelled picture with the matching explanation. Open the vector diagram

Why this result follows

What this example does not establish

MAIN LESSON · SECTION 7

Audit the reference, information access and complete procedure

> A useful baseline is a real alternative under the intended task constraints.

Core fairness qualification and failure analysis

A **baseline** is credible relative to the task, not merely a procedure easy to outperform.

A useful check asks what would make the reasoning fail. Distinguish a failure to execute the declared procedure from a weakness in the procedure itself.

The first may require a correction; the second requires an explicitly revised method and new **evidence**. Preserve unsuccessful cases so an improved result cannot hide the original problem.

Complete source, leakage and resource probes

Fair-baseline probe	Expected interpretation
Baseline intentionally random despite available credible lookup	Weak reference can exaggerate apparent improvement
B current sources, A obsolete sources	Information freshness confounded with method
Both use current permitted register	Appropriate matched task evidence; not automatically all other controls
Only B sees hidden judge labels	Evaluation leakage; not an unseen fair test
Same headline score but resources unknown	Cannot infer equivalent cost or budget performance
Higher quality violates task time budget	Report complete procedure trade-off; no budget-feasible superiority claim
Small aggregate score gap	Descriptive result; stable advantage/significance need appropriate evidence
Unknown paired case outcomes	Net correct-count change does not identify every gain/loss
Conditions audit has no recorded mismatch	Checks declared metadata only, not actual execution verification

Guided repair and explanation

- A report omits this qualification: A **baseline** is credible relative to the task, not merely a procedure easy to outperform. Explain how omitting it could change a reader's interpretation, and write an accurate replacement.

Guided: restate the worked result, then name the assumption or limitation that prevents a broader conclusion.

State one repair and the checks you would rerun.

▼ **Earlier wording, alternatives and picture transcripts**

Locate the earliest unsupported step

MAIN LESSON · SECTION 8

Practise identifying the exact comparison mismatch

> Name what must be aligned or disclosed before claiming a method improvement.

Current explanation and guided check

- Explain the weekly **mechanism** in your own words. Use the worked scenario to name its **input**, decision rule and **output**.

Guided: restate the worked result, then name the assumption or limitation that prevents a broader conclusion.

Explain your prediction to a partner before comparing results.

Supported W2 trace

Use workbook W2 for the connected example "**Baseline**s and fair comparisons". First fill the **input** and rule boxes. Try the middle steps before asking for a hint. After a hint, try the next step yourself.

Plan	Do	Check
What is given? Which rule or procedure applies?	Record each intermediate step.	Does the result follow? What remains unknown?

Explain your trace to a partner. Your partner points to one step to check. Swap roles on the next case. Each person writes their own explanation.

Individual scope check

- Use the worked case for **Baseline**s and fair comparisons. Proposed explanation: "The worked result establishes every future case". Decide whether this explanation is supported.

  Point to the step or supplied fact that decides; explain your repair if needed.

- Answer on your own before discussion. If you are unsure, say which step you need help with.

Picture response

- Calculate 6/8 and 7/8 correctness and name the one-case gain. Why is giving B the answer key incompatible with an unseen comparison to the simple **baseline**?

My explanation using a pictured step and the given facts

Reserve the selected per-case time-budget scenario

This is supported practice. Keep the separately named independent case for an attempt before feedback.

▼ Earlier wording, alternatives and picture transcripts

Practise with less help: Trace the worked case

Check your reasoning

Explain the picture

MAIN LESSON · SECTION 9

Run a complete source-freshness counterexample and comparison audit

> Inspect the known mismatches and the conditions still unreported.

Current old/current source comparison investigation

Assistant gets current register; **baseline** gets an obsolete one. Assistant scores higher. Show the intermediate steps and give the conclusion with its scope.

Runnable constructed lookup and condition audit

```
"""Constructed same-procedure/different-source comparison, not a model benchmark."""
from fractions import Fraction
from copy import deepcopy
CURRENT={'V':'E','W':'B','X':'D'}
OLD={'V':'C','W':'B','X':'A'}
CASES=['V','W','X'];TRUTH=deepcopy(CURRENT)

def evaluate(source,cases,truth):
 if type(source) is not dict or type(cases) is not list or not cases or len(set(cases))!=len(cases):
  raise ValueError('supplied source and unique matched cases required')
 if set(cases)!=set(truth):raise ValueError('complete fixed judged cases required')
 outputs=[{'case':case,'answer':source.get(case),'correct':source.get(case)==truth[case]} for case in
cases]
 return {'outputs':outputs,'correct':sum(row['correct'] for row in outputs),'count':len(cases)}
# A and B intentionally use the identical lookup function; only information changes.
a_old=evaluate(OLD,CASES,TRUTH);b_current=evaluate(CURRENT,CASES,TRUTH)
a_aligned=evaluate(CURRENT,CASES,TRUTH)
assert a_old['correct']==1 and b_current['correct']==a_aligned['correct']==3

def conditions(case_ids,source_version,*,test_labels_seen=False):
```

```
 return {'case_ids':list(case_ids),'source_version':source_version,'scoring':'fixed-rubric-v1',
         'test_labels_seen':test_labels_seen,'budget_policy':'not supplied'}
def compare_conditions(a,b):
 required={'case_ids','source_version','scoring','test_labels_seen','budget_policy'}
 if type(a) is not dict or type(b) is not dict or set(a)!=required or set(b)!=required:
  raise ValueError('complete declared condition records required')
 if type(a['test_labels_seen']) is not bool or type(b['test_labels_seen']) is not bool:
  raise ValueError('explicit test-label exposure flags required')
 issues=[]
 for name in ['case_ids','source_version','scoring']:
  if a[name]!=b[name]:issues.append('mismatched '+name)
 if a['test_labels_seen'] or b['test_labels_seen']:issues.append('test answer-key exposure')
 if a['budget_policy']=='not supplied' or b['budget_policy']=='not supplied':issues.append('resource policy
not supplied')
 elif a['budget_policy']!=b['budget_policy']:issues.append('mismatched resource policy')
 return issues
unequal=compare_conditions(conditions(CASES,'old'),conditions(CASES,'current'))
assert 'mismatched source_version' in unequal
# Separate picture: aggregate correctness, same eight cases/sources, no tuning on test.
picture_a=Fraction(6,8);picture_b=Fraction(7,8)
assert 100*picture_a==75 and 100*picture_b==Fraction(175,2)
assert 100*(picture_b-picture_a)==Fraction(25,2)
picture_ids=['P'+str(i) for i in range(1,9)]
leaked=compare_conditions(conditions(picture_ids,'same'),conditions(picture_ids,'same',test_labels_seen=True))
assert 'test answer-key exposure' in leaked
print('Added same-procedure old/current source results:',a_old,b_current)
print('Align both to current sources:',a_aligned,b_current,'not a claim about real model equality')
print('Declared-condition issues:',unequal)
print('Separate picture percentages:',float(100*picture_a),float(100*picture_b),'net count difference: one;
paired flips unknown')
print('Picture answer-key-exposure variant issues:',leaked)
print('Current task evidence is permitted source material, distinct from hidden evaluation labels. These
local fixtures do not measure runtime budgets, real model quality, statistical significance or broad
advantage.')
```

Expected results and execution/measurement limits

1. Predict each three-case old/current lookup and score. The same deterministic function gives 1/3 on the old register and 3/3 on current; using current for both gives 3/3 each. This constructed counterexample shows how a score gap can arise without a procedure difference, not what a real model would score. The condition checker records source mismatch and unprovided resource policy.

   Separate picture arithmetic computes 75%/87.5% and marks answer-key exposure as a violation. It uses declared condition records, not proof of unseen execution, actual runtime budgets or model quality. Keep the independent twelve/two/three-second scenario reserved.

Evidence ledger and reflection

Record	Your evidence
Given input and fixed rule	
Prediction before checking	
Actual result, or “not run”	
First mismatch and one explanation	
Boundary or missing input and expected response	

On paper, trace the supplied procedure. If code is provided, predict before running it in a local practice file. Never write an expected answer as though it were an observed run.

Before trying: what do I expect, and why? While trying: which step could first go wrong? Afterwards: what did I actually observe, and what should change? Keep “not run” if you only traced the procedure.

▼ **Earlier wording, alternatives and picture transcripts**

Predict, investigate and preserve the evidence

Plan → try → check

MAIN LESSON · SECTION 14

Try alone: a more expensive model under a strict task budget

> Compare complete procedures rather than quality alone.

Complete selected independent task

Model uses 12 seconds and **baseline** 2 on each case. Task has a 3-second budget. How should results be framed?

First attempt and support record

My independent explanation, trace and limits

Attempt this case before feedback. Record any help. Earlier examples below are further practice; a case already practised with feedback cannot establish fresh transfer.

Feedback after trying

Attempt the independent case before hints. Record whether you worked alone, used a hint or followed a model. After feedback, fix the first unsupported step and explain why it changed.

A later check uses a changed case, not a copied answer.

▼ **Earlier wording, alternatives and picture transcripts**

Independent case: Apply a new case independently

Attempt this without hints

Model uses 12 seconds and **baseline** 2 on each case. Task has a 3-second budget. How should results be framed?

My independent procedure, result and limitation

Record support used: none, hint or teacher explanation. After feedback, a different case is needed to demonstrate independence.

Use feedback to improve

Save the baseline implementation and every comparison condition

> Keep permitted sources, hidden labels and resource budgets explicit.

Project task and artifact

Add a labelled **baseline**s and fair comparisons **evidence** sheet to Reproducible AI Research Study. Show how you can explain the **mechanism** and terminology of **baseline**s and fair comparisons.

Use the supplied fictional case and keep its **input**s, intermediate steps, calculation or **prediction**, and limitation. Connect this record to last week's record; a matching answer without a trace is insufficient.

Project evidence	Record
Component and version	
Inputs permitted at prediction time	
Trace and comparison with earlier work	
Failure, limit and human review	

Actual local calculation or paper trace and limitations

For the Reproducible AI Research Study, retain an annotated worked trace and the changed-case reasoning. Connect the claim to its records, units and assumptions so another investigator can challenge or reproduce it.

▼ **Earlier wording, alternatives and picture transcripts**

Evidence for your project

MAIN LESSON · SECTION 16

Explain the observed comparison and the constraint it does not settle

> A better score need not be a fair or feasible task improvement.

Retrieve and explain

- Close your notes. Explain this goal in your own words: Explain the **mechanism** and terminology of **baseline**s and fair comparisons

> **My deciding step, evidence and one remaining question**
>
> | |
> |---|
> | |
> | |
> | |
> | |

Plan a changed-case return

If you are unsure, return to the worked picture and explain one arrow. If you are ready, change an **input** or assumption and justify your **prediction**.

Save your workbook **evidence**. At the next review, try a different case before opening your old answer.

Reference bank and extra practice

Choose the reference that helps your next step.

▼ **Reference · Use research vocabulary precisely**

REFERENCE · SECTION 5

Reference · Use research vocabulary precisely

> Connect each term to the question, comparison, measurement or evidence.

Vocabulary and your explanation

Word	Meaning
Baseline	A relevant reference procedure
Matched comparison	A comparison with aligned task conditions
Resource budget	The permitted time or computation

Use two terms to explain a step in this week's example.

▼ **Extra challenge · Explain a stronger case**

REFERENCE · SECTION 10

Extra challenge · Explain a stronger case

> Choose the next challenge after the main route.

Explain,repair and extend

- Explain the weekly **mechanism** in your own words. Use the worked scenario to name its **input**, decision rule and **output**.

- A report omits this qualification: A **baseline** is credible relative to the task, not merely a procedure easy to outperform. Explain how omitting it could change a reader's interpretation, and write an accurate replacement.

- For the worked scenario, propose one additional case that could reveal a failure, describe its expected outcome before running it, and explain what would need to remain fixed for a fair comparison.

  Do not reuse W4 as a new final test after receiving feedback.

Explain → apply → challenge

> **Explain which evidence supports your conclusion and which claim remains unproved.**

▼ **Reference · Check the research reasoning and its limits**

REFERENCE · SECTION 12

Reference · Check the research reasoning and its limits

> Use after practice; reserve the selected independent case for its first attempt.

Current explanation and response

- Complete the supported workbook tasks before attempting W4 independently. Record any hints or help used. After feedback, use the teacher's fresh retry rather than copying the earlier answer.

  For a project, select a relevant new case, predict the outcome from the declared rule and preserve the result alongside any revision.

- For the worked scenario, propose one additional case that could reveal a failure, describe its expected outcome before running it, and explain what would need to remain fixed for a fair comparison.

  Do not reuse W4 as a new final test after receiving feedback.

> **Exit: propose one revealing change to the worked scenario. State the expected outcome and the evidence that would make you revise your explanation.**

> **My limited claim and a useful next check**

▼ **Extra practice · Revisit the complete trace**

REFERENCE · SECTION 13

Extra practice · Revisit the complete trace

> Predict before reading the supplied steps.

Complete trace and evidence

Complete evidence record

Baselines and fair comparisons — evidence trace

INPUT AND SCOPE	CHECK OR CALCULATION	CONCLUSION AND LIMIT
Assistant gets current register; baseline gets an obsolete one. Assistant scores higher.	For a room-reporting study use manual authoritative lookup or a deterministic lookup matching the requested room, not random room names.	Source freshness is confounded with procedure. Provide both the same eligible current sources or report the unequal comparison without isolating method.

The figure connects this chapter's worked input, declared procedure and result. Read the complete worked reasoning for qualifiers and intermediate steps; the abbreviated labels do not replace that reasoning.

Follow the **input** through the stated procedure to its result. Cover the result first, predict it, then explain every intermediate step. The text/table accompanies the figure so the example can be read without colour.

> **My complete trace of the worked case**

▼ **Reference · The earlier route checklist**

REFERENCE · SECTION 15

Reference · The earlier route checklist

> Use one main route rather than repeating the lesson.

Targets,route and first prediction

Start with this question: A **baseline** deserves a fair chance. Two classifiers answer the same eight fixed questions with the same allowed sources.

By the end, I can	How I will show it
Explain the mechanism and terminology of baselines and fair comparisons	Explain the deciding step, show a trace or test, and state one limit.
Apply the stated procedure to a new case with a traceable result	Explain the deciding step, show a trace or test, and state one limit.
Identify an assumption or limitation and design a revealing follow-up	Explain the deciding step, show a trace or test, and state one limit.

Remember → watch and explain → try with help → investigate → try independently → keep project **evidence**. Your teacher may spread this lesson across several classes.

Keep your first **prediction**. Compare it with what the **evidence** shows and explain any change.

▼ **Extra practice · Predict,check and repair**

REFERENCE · SECTION 17

Extra practice · Predict,check and repair

> Keep supported feedback separate from the independent case.

First supported question and optional hint

- A **baseline** deserves a fair chance. Two classifiers answer the same eight fixed questions with the same allowed sources.
- Which labelled step matches this detail? B adds one correct case here; compare resources and failure types too.
- 1. Declare **baseline**
- 2. Interpret cautiously

- 3. Measure fairly

- Hint if needed: Read the steps in the pictorial case. Match the supplied detail to the stage that performs or records it.

Practice 1: explain the deciding evidence

My selected conclusion, deciding fact and explanation

▼ **Check after the first attempt**

After your attempt, compare with the supported explanation: Interpret cautiously B adds one correct case here; compare resources and failure types too. A **baseline** anchors the claim of improvement.

Fair comparison keeps information access, cases and scoring aligned while allowing the declared method difference.

Changed-case question and optional hint

- Changed-case practice: Give B the answer key before testing. Which conclusion is supported by the supplied facts?

- 1. We can decide the result without examining the changed **input** or the procedure.

- 2. The resulting comparison is invalid as unseen evaluation.

- 3. This one example guarantees correct results for every future **input**.

- Hint if needed: Use the changed condition and the deciding step in the pictorial trace. The answer must say what follows from those facts.

Practice 2: explain the deciding evidence

My selected conclusion, deciding fact and explanation

▼ Check after the changed attempt

After your attempt, compare with the supported explanation: The resulting comparison is invalid as unseen evaluation. B would receive privileged information that A did not receive.

Repair and retain the record

Review Hypotheses and measurable outcomes (ai-7/7). Goal: Explain the **mechanism** and terminology of hypotheses and measurable outcomes. Short practice: Name the predicted outcome and how unsupported is scored.

If B sounds friendlier but ties A's unsupported count, does the recorded improvement hypothesis receive support?. Return to this week and explain what you changed.

Review Replication and reproduction (ai-7/5). Goal: Explain the **mechanism** and terminology of replication and reproduction. Short practice: Calculate the scores from the original and new four-record sets.

Which activity reproduces saved analysis, which repeats the protocol, and why does a different rerun of the original data need investigation?. Return to this week and explain what you changed.

Review Reading a research claim (ai-7/2). Goal: Explain the **mechanism** and terminology of reading a research claim. Short practice: Calculate the **baseline** and new rates for 2/10 and 4/10.

Explain both twice the rate and the percentage-point change, and say why neither measures every learner's improvement.. Return to this week and explain what you changed.

Use Download completed work in the hosted lesson to export your answers, reflections, practice feedback and code-run **evidence**. Open the HTML copy to read or print it.

Drafts stay in this browser when storage is available; clear drafts on a shared device.

▼ Earlier wording, alternatives and picture transcripts

Use this supported practice once, in the browser or on paper. Keep the reserved independent assessment for a separate first attempt before teacher feedback.

Repair a missing building block

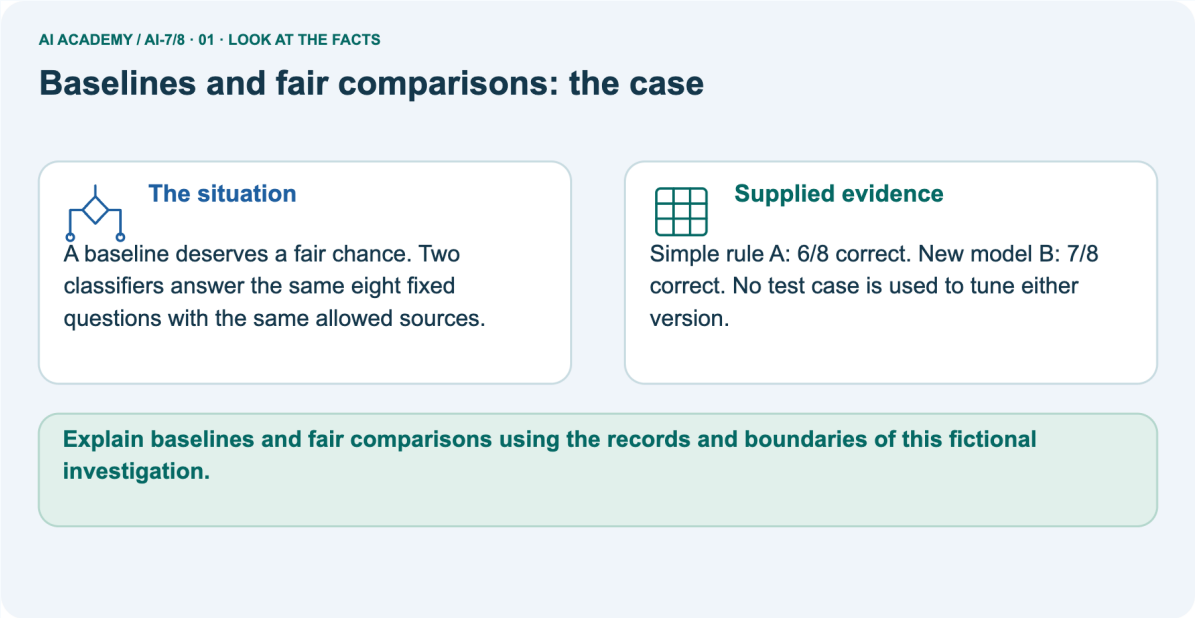
Keep your completed work

▼ Reference · Six-picture concept journey

Reference · Six-picture concept journey

> Choose the picture that supports the current explanation.

01. Baselines and fair comparisons: the case



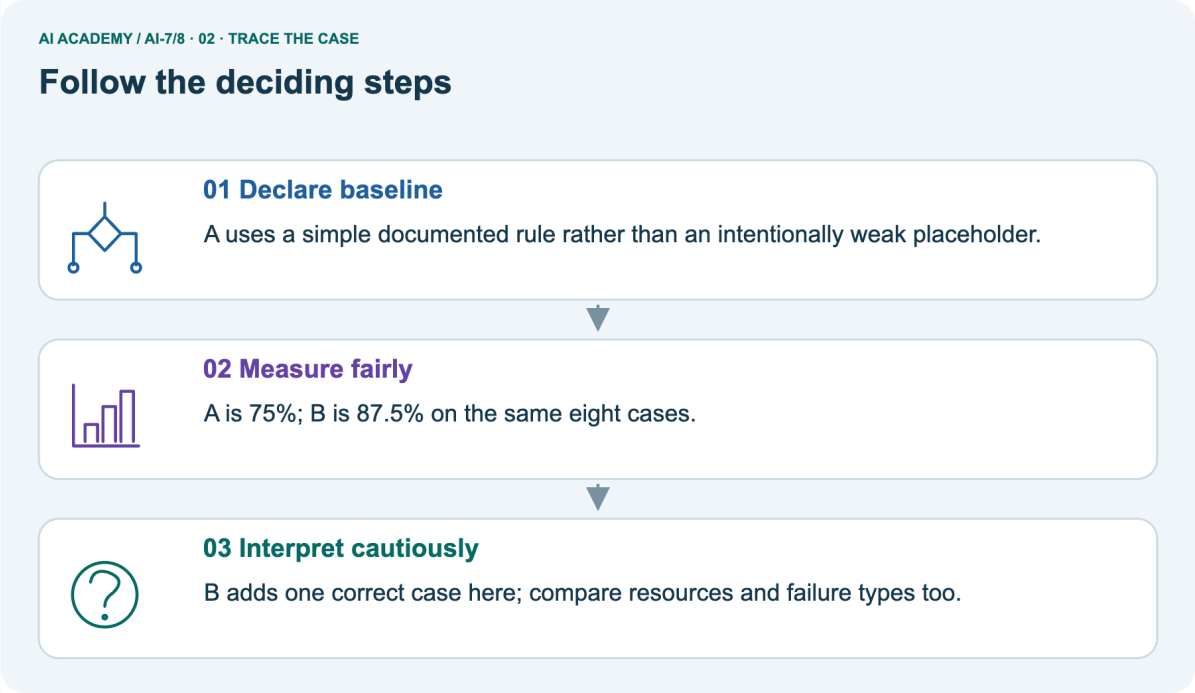
Follow the labelled picture with the matching explanation. Open the vector diagram

A **baseline** deserves a fair chance. Two classifiers answer the same eight fixed questions with the same allowed sources. Supplied facts: Simple rule A: 6/8 correct. New model B: 7/8 correct.

No test case is used to tune either version.

Point to the given fact that matters most. Explain why it matters.

02. Follow the deciding steps

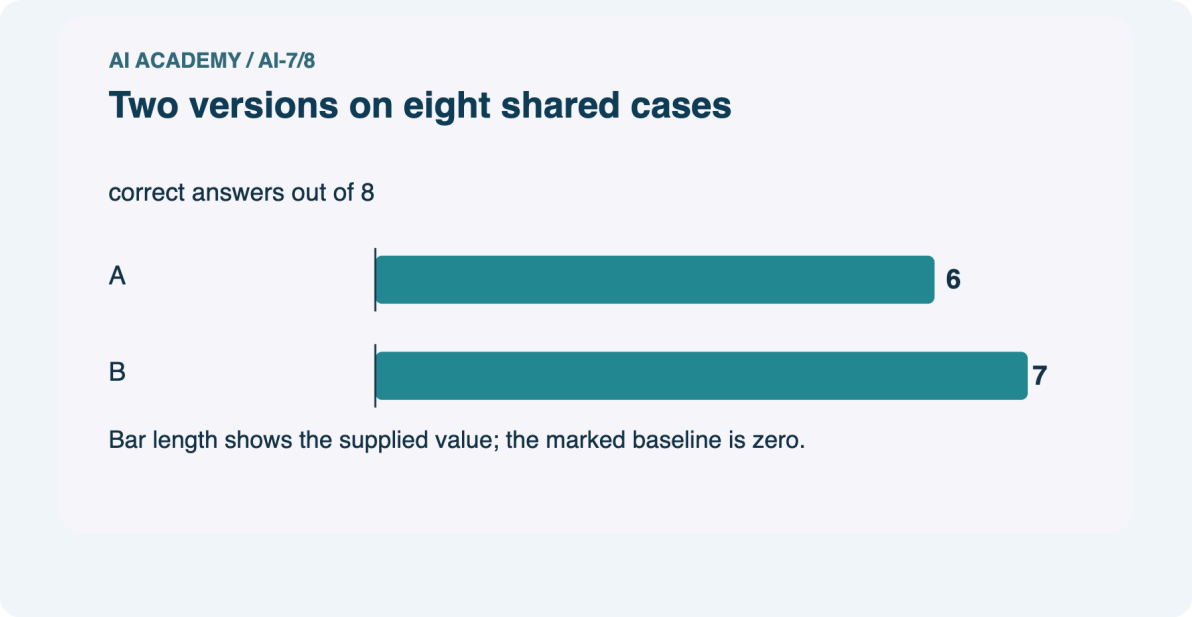


Follow the labelled picture with the matching explanation. Open the vector diagram

Declare **baseline**: A uses a simple documented rule rather than an intentionally weak placeholder. → Measure fairly: A is 75%; B is 87.5% on the same eight cases. → Interpret cautiously: B adds one correct case here; compare resources and failure types too.

Point to each numbered stage and explain which recorded **input** it uses.

03. Two versions on eight shared cases

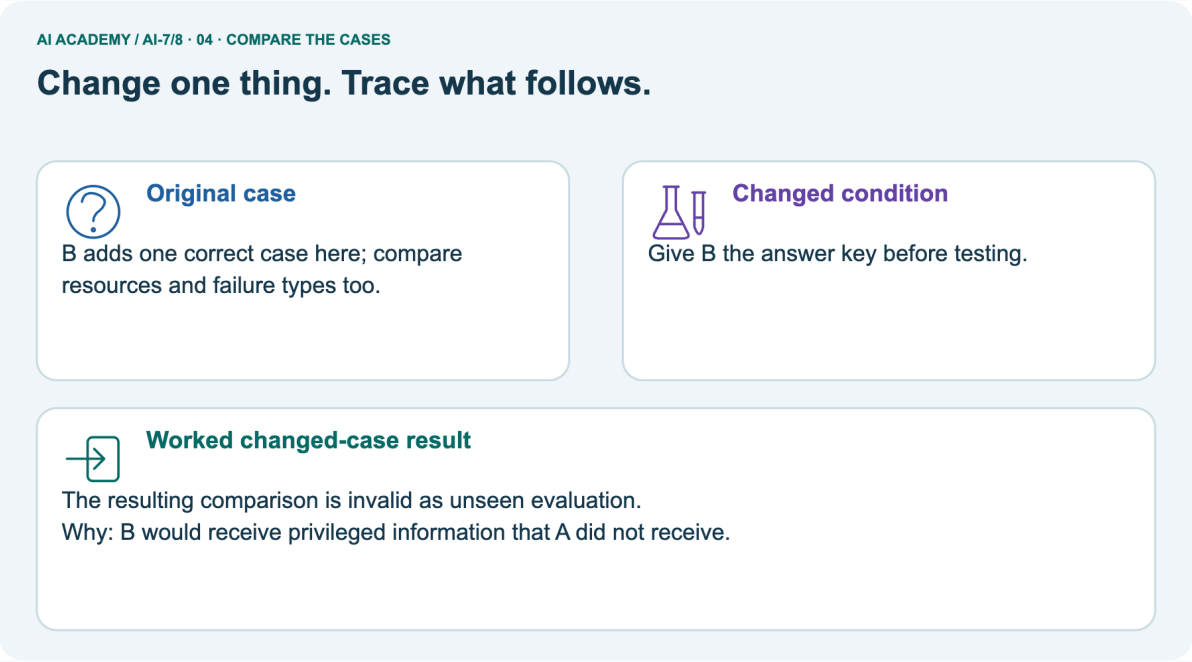


Follow the labelled picture with the matching explanation. Open the vector diagram

Two versions on eight shared cases. Units: correct answers out of 8. A: 6. B: 7.

Read the labels and values. Explain the relationship the model shows.

04. Change one thing. Trace what follows.



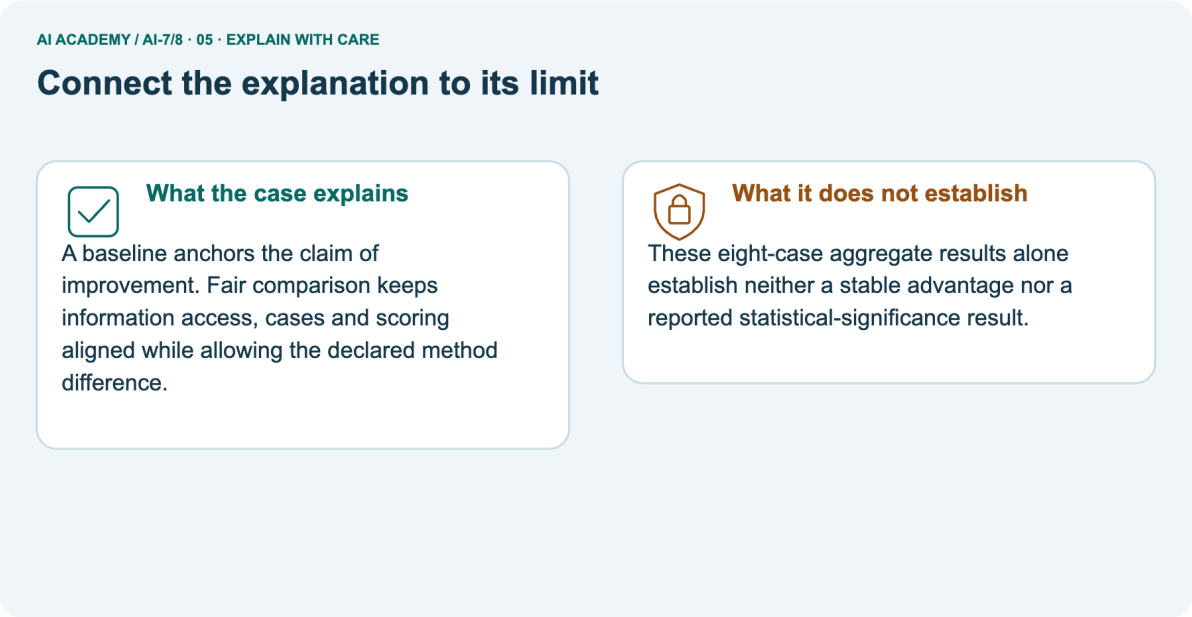
Follow the labelled picture with the matching explanation. Open the vector diagram

Original case: B adds one correct case here; compare resources and failure types too. Changed condition: Give B the answer key before testing. Expected result: The resulting comparison is invalid as unseen evaluation.

Reason: B would receive privileged information that A did not receive.

Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude.

05. Connect the explanation to its limit

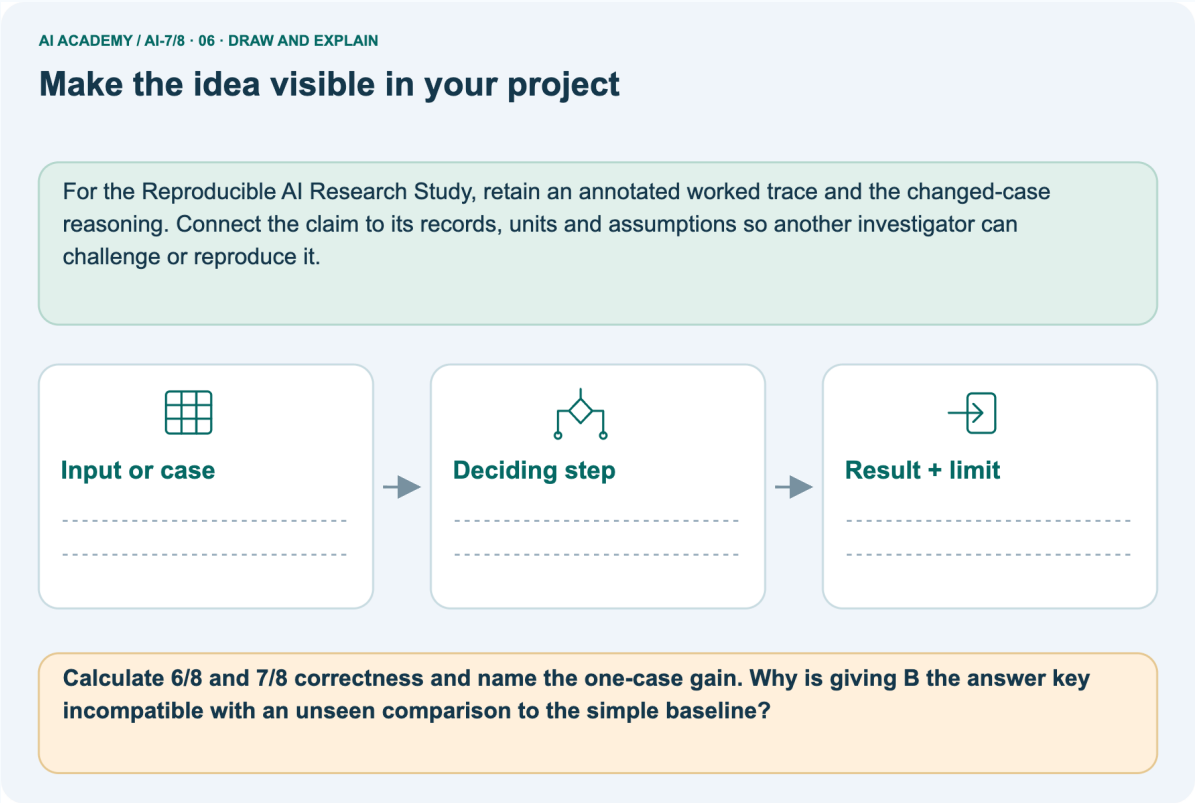


Follow the labelled picture with the matching explanation. Open the vector diagram

A **baseline** anchors the claim of improvement. Fair comparison keeps information access, cases and scoring aligned while allowing the declared method difference. Limit: These eight-case aggregate results alone establish neither a stable advantage nor a reported statistical-significance result.

Explain the case, then point to a limit on the claim.

06. Make the idea visible in your project



Follow the labelled picture with the matching explanation. Open the vector diagram

For the Reproducible AI Research Study, retain an annotated worked trace and the changed-case reasoning. Connect the claim to its records, units and assumptions so another investigator can challenge or reproduce it.

Draw your **input**, deciding step, result and limit. Supported question: Calculate 6/8 and 7/8 correctness and name the one-case gain. Why is giving B the answer key incompatible with an unseen comparison to the simple **baseline**?

Draw your project trace and justify the deciding step using **evidence**.

Earlier wording, alternatives and picture transcripts

01. Baselines and fair comparisons: the case

02. Follow the deciding steps

03. Two versions on eight shared cases

04. Change one thing. Trace what follows.

05. Connect the explanation to its limit

06. Make the idea visible in your project

Optional interactive practice and completed work

Save or print your responses for your portfolio. Progress and drafts on the hosted website stay in the current browser. Use only fictional or teacher-approved data.